

11.5 Writing Exponential Functions From a Graph or Table

Lesson Introduction

 Benchmark	MA.912.AR.5.4 Write an exponential function to represent a relationship between two quantities from a graph, a written description, or a table of values within a mathematical or real-world context.		 Learning Targets	- I can write an exponential function from a graph. - I can write an exponential function from a table.
 Benchmark Coherence	Building On N/A		Working Toward MA.912.AR.5.5 Given an expression or equation representing an exponential function, reveal the constant percent rate of change per unit interval using the properties of exponents. Interpret the constant percent rate of change in terms of a real-world context.	
 Essential Questions	- How can you write an exponential function given a graph? - How can you write an exponential function given a table?			
 Lesson Narrative	In this lesson, students continue working with exponential functions. Students will begin this lesson by learning how to write an exponential function when its graph is provided. They will learn the two different exponential function forms and how to leverage both in a given situation. They will then transition to being presented with an exponential function that is represented as a table of values and learn how to write a function from the table.			
 Component Summary	11.5.1 Warm-Up (~5 min.)	Growth mindset reflection (<i>SE p. 181</i>)	11.5.3 Guided Practice (15 min.)	Practice writing an exponential function given a graph or a table (<i>SE p. 184</i>)
	11.5.2 Guided Instruction (20 min.)	Introduction to writing an exponential function given a graph or a table (<i>SE p. 182</i>)	11.5.4 Wrap-Up (~5 min.)	Ticket Out the Door writing an exponential function given a graph and a table (<i>SE p. 185</i>)
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> N/A		N/A	Work through each of the problems in this lesson prior to facilitating it. Note places where you want to do checks for understanding before moving forward.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may forget how to start writing the function from a graph (finding the value of a first). - Students may not connect finding the a value from a graph with finding the a value from a table.	- This lesson contains content that can generally be a struggle for most students. The structure of the entire lesson is guided, as you will want to ensure you are able to assist students step by step through this process. - Consider opportunities for making connections to the concepts students learned in the first four lessons that will help them as they determine how to write the equations from given information. - It will be critical to use explicit and consistent language during instruction.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Compare and Connect 11.5.2 Guided Instruction provides students the opportunity to receive guided instruction on writing exponential functions from graphs and tables. The language and precise structure of this process is critical for student success. During the instruction component, students should be able to hear your thinking out loud to connect what you're doing to what you're thinking and processing.	MA.K12.MTR.2.1: Multiple Representations In this lesson, students will be given an exponential function represented as a graph or table of values. They will use what they learn in this lesson to write the equations of the function in different forms. The lesson centers around multiple representations and how to relate each representation to the other.
 Differentiate Instruction	In 11.5.2 Guided Instruction, consider creating the graph that goes with the table of values in Question 3 and putting them side by side in a central location where all students can see it (i.e., an anchor chart). This can help students make connections between the two representations. 11.5.4 Wrap-Up has students complete a Ticket Out the Door to demonstrate their understanding of writing exponential functions given a table or a graph. Student performance on this task can provide information on which form students may still need additional support for in Lesson 6.	
Lesson Notes:		

11.5.1 Warm-Up: Growth Mindset

Component Overview: Students will complete a growth mindset activity where they reflect on making mistakes.



11.5.1 Warm-Up: Growth Mindset

1. It's crucial to create an environment where failure, errors, and mistakes are seen as a natural part of learning. Making that environment all starts with you; more specifically, your self-talk.

- a. When you make a mistake, what do you say to yourself?

Answers will vary.

- b. Explain if what you say to yourself reflects a growth or fixed mindset.

Answers will vary.



When reviewing this activity, consider creating a t-chart that has “Growth Mindset” on one side and “Fixed Mindset” on the other. As students share out examples, have the class determine which side of the chart the statements should go on.

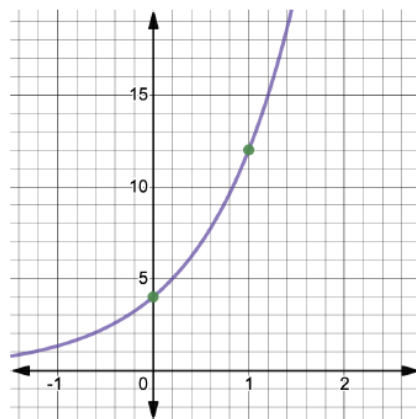
11.5.2 Guided Instruction: Writing an Exponential Function From Graphs and Tables

Component Overview: Students will be introduced to writing exponential functions given a table or a graph. In the first two problems, students are working with a given graph. In question 3, students are working with a given table of values. The last question has students practice with both a table and a graph.



11.5.2 Guided Instruction: Writing an Exponential Function from Graphs and Tables

1. The graph of an exponential function is shown.



To express this exponential function in the form $f(x) = a(b)^x$ or $f(x) = a(1 \pm r)^x$, the values for a and b need to be found. These values can be found using two points on the graph and the equation in the form $y = a(b)^x$.

- a. Find the value of a by substituting point $(0, 4)$ in for x and y in $y = a(b)^x$.

$$\boxed{4} = a(b)^{\boxed{0}}$$

$$a = 4$$

- b. Discuss with your partner if the point $(1, 12)$ can be used to determine the value of b . Write a brief summary of your discussion.
Yes, $(1, 12)$ can be used to determine the value of b , because it is a point on the exponential function since we know the value of a from the earlier question.



When beginning this activity, consider leveraging a “brain dump” activity regarding exponential functions. Ask students to tell you everything they recall about exponential functions from what they’ve learned in this unit. As students share, consider charting that information up on the board for all students to see. Once completed, ask students to look at the graph in question 1 to see what they know about it based on all the things they’ve already learned in this unit from the information that was just charted.

Then introduce students to the two forms of exponential functions before guiding them through question 1.



Why do you think using the point $(0, 4)$ is an efficient way to find the value of a ?



Explain whether there are other points that could be used to determine the value of b .

11.5.2 Guided Instruction: Writing an Exponential Function From Graphs and Tables (continued)

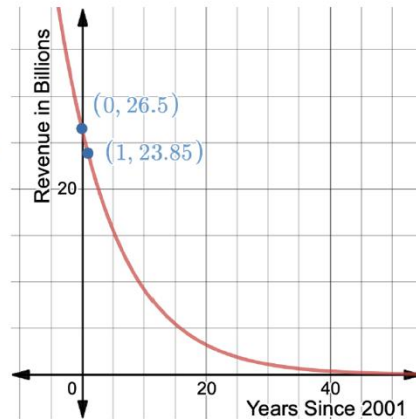
- c. Use the value of a found in Part A and the point $(1, 12)$ to determine the value of b .

$$b = 3 \quad \boxed{12} = \boxed{4} (b)^{\boxed{1}}$$

- d. Write the function for the graph using both forms.

$f(x) = a(b)^x$	$f(x) = a(1 \pm r)^x$
$f(x) = 4(3)^x$	$f(x) = 4(1 + 2)^x$

2. The Global Music Report shows there has been a decline in revenue from physical records due to the increase of digital downloads and streaming services. The graph shows y , the revenue in billions, and x , years since 2001.



- Discuss with your partner which point can be used to find the a value for the equation $y = a(b)^x$.
 $(0, 26.5)$
- Work with your partner to write the equation for the revenue of physical records since 2001.
 $y = 26.5(0.9)^x$
- What does the a value represent in the context?
The company revenue in 2001.
- What does the b value represent in the context?
The revenue's rate of decay each year.



Be very clear in the language used in question 2c. It's critical for students to understand how this step is different from what they did in question 2a. Consider writing out the concept of what is occurring and then the method used to do so.

Example:

- Step 1: Find the value of a
 - Use $f(x) = a(b)^x$ by plugging in the values of the y -intercept and solving for a .
- Step 2: Find the value of b
 - Use $f(x) = a(b)^x$ by plugging in the values of another point and the value determined for a . Then solve for b .



Ask students about what they know about this graph and what it could mean in the context of the problem. Consider using the mark-up strategy to note connections between the two representations.



For an additional challenge, have students determine what the point $(1, 23.85)$ means in the context of this problem.

11.5.2 Guided Instruction: Writing an Exponential Function From Graphs and Tables (continued)

3. Writing an equation from a table of values can be done by finding the initial value, a , and the common ratio, b . The table of values represents an exponential relationship.

x	0	1	2	3	4	5	6
$f(x)$	$\frac{2}{3}$	2	6	18	54	162	486

- a. Describe how the values of $f(x)$ change as the x -values increase by 1.

They triple, or are multiplied by 3.

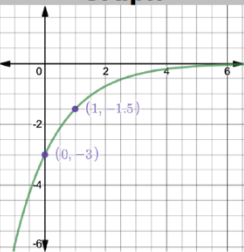
- b. What is the initial value, a ?

$\frac{2}{3}$

- c. Discuss with your partner how to use the information determined in Parts A and B to write the function in the form $f(x) = a(b)^x$. Then, write the function.

$$f(x) = \frac{2}{3}(3)^x$$

4. When the value of a is negative, the same process can be used. Write an exponential function for the graph and for the table of values in the form $f(x) = a(b)^x$.

Graph	Table												
	<table> <tr> <th>x</th><th>$f(x)$</th></tr> <tr> <td>0</td><td>-2</td></tr> <tr> <td>1</td><td>-7</td></tr> <tr> <td>2</td><td>-24.5</td></tr> <tr> <td>3</td><td>-85.75</td></tr> <tr> <td>4</td><td>-300.125</td></tr> </table>	x	$f(x)$	0	-2	1	-7	2	-24.5	3	-85.75	4	-300.125
x	$f(x)$												
0	-2												
1	-7												
2	-24.5												
3	-85.75												
4	-300.125												
Function	Function												
$f(x) = -3(0.5)^x$	$f(x) = -2\left(\frac{7}{2}\right)^x$												



Model for students how to use a table of values to determine the values of a and b to write the equation of the function.

Students may not be familiar with the term “common ratio” in question 3. This may be something you want to clarify as part of this process.



- How do the values in questions 3a and 3b relate to the process that was used when a graph was provided?*
- Which representation do you find efficient to write the function from?*



Consider having students work with a partner to attempt question 4 together and then review as a class. Use this time to work with students who you observed struggling during the first three questions to clear up any misconceptions and clarify the intended learning.



For students that finish early, ask them to determine the impact on the graph if the a term from question 1 was negative.

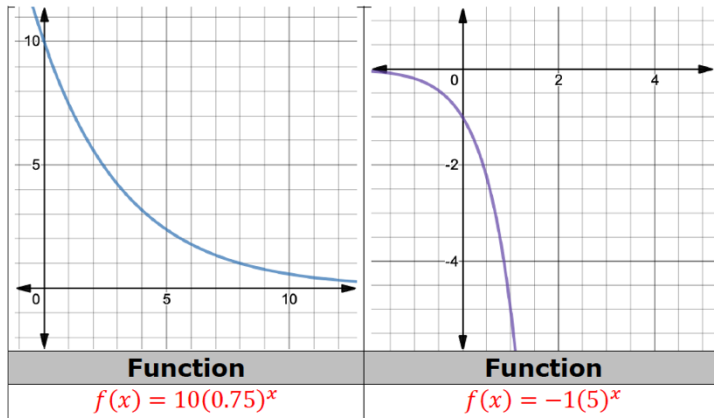
11.5.3 Guided Practice: Writing Exponential Functions

Component Overview: Students will practice writing exponential functions from a graph and then from a table.



11.5.3 Guided Practice: Writing Exponential Functions

1. Write the exponential function for each graph.



2. Write an exponential function for each table of values.

x	$f(x)$	x	$f(x)$
0	4	0	20
1	10	1	16
2	25	2	$\frac{64}{5}$
3	62.5	3	$\frac{256}{25}$
4	156.25	4	$\frac{1,024}{125}$
Function		Function	
$f(x) = 4\left(\frac{5}{2}\right)^x$		$f(x) = 20\left(\frac{4}{5}\right)^x$	



Use informal observation from the previous component to determine how “guided” this Guided Practice should be for you and your classroom. This could mean small-group instruction for some, while others attempt independently, or it could mean teacher facilitation of practice with students actively working as a whole group.



Without writing the function yet, what can you tell about the difference in the values of a for each of the graphs?



Students may get uncomfortable with these tables because they see decimal and fractional values. Encourage students to continue using the process they’ve learned in this lesson. Watch for students who are completing the steps correctly but are making computation errors versus students who are not able to write the function correctly at all.

11.5.4 Wrap-Up: Ticket Out the Door

Component Overview: Students will complete a Ticket Out the Door that assesses their level of mastery with writing exponential functions given a graph or a table. Students should be allowed to access a calculator to complete the Wrap-Up.



11.5.4 Wrap-Up: Ticket Out the Door

- Write the exponential function represented by the graph and the exponential function represented in the table of values.

Graph	Table												
	<table> <tr> <th>x</th><th>$f(x)$</th></tr> <tr> <td>0</td><td>0.7</td></tr> <tr> <td>1</td><td>3.5</td></tr> <tr> <td>2</td><td>17.5</td></tr> <tr> <td>3</td><td>87.5</td></tr> <tr> <td>4</td><td>437.5</td></tr> </table>	x	$f(x)$	0	0.7	1	3.5	2	17.5	3	87.5	4	437.5
x	$f(x)$												
0	0.7												
1	3.5												
2	17.5												
3	87.5												
4	437.5												
Function	Function												
$f(x) = 18\left(\frac{1}{3}\right)^x$	$f(x) = 0.7(5)^x$												



Student performance on this activity can provide data on which representation, if any, students may need additional support with.